

Three framework-specification questions for a first-principles (r_e) derivation

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Three framework-specification questions for first-principles r_e derivation — for Tepper Gill

Date: 2026-05-26 (PM, after the day's overnight progress runs). **From:** Trey Morris (with Claude Opus 4.7). **Re:** Follow-up to the earlier candidates note¹ and triangulation note²; tracks against the framework-specifications issue³ under the master derivation thread⁴.

We present here three framework-specification questions that have proved necessary across the three parallel candidate investigations of a first-principles derivation of r_e/r_0 . Each question, if resolved, would permit a derivation that goes beyond the algebraic inheritance from QED's $a_e(\alpha)$ that the present closed-form result expresses.

What the three investigations converged on

Candidate 3 confirmed, by a line-by-line reading of §III.D Eqs. (III.18)–(III.23)⁵, that the as-published apparatus leaves r_e/r_0 as a free parameter. The algebraic-back-fit match to the closed form

$$\frac{r_e}{r_0} = \frac{2 - \alpha/(2\pi)}{4 + \alpha/\pi} \quad (1)$$

is necessary but not sufficient for intentional Schwinger encoding. A cross-particle test against the FNAL Muon $g - 2$ 2023 measurement, performed in the same investigation, yielded

$$\frac{r_\mu}{r_0^\mu} - \frac{r_e}{r_0^e} = 3.13 \times 10^{-6} \quad (2)$$

¹Candidates note: [Roadmapping/Author_Reports/2026-05_re_derivation_candidates_for_gill.md](#).

²Triangulation note: [Roadmapping/Author_Reports/2026-05_re_triangular_followup_for_gill.md](#).

³Issue #75 (framework-specifications), [github.com/temoTxt/PyPhysics/issues/75](#).

⁴Master derivation thread, issue #67, [github.com/temoTxt/PyPhysics/issues/67](#).

⁵Candidate 3: issue #66 and PR #70, [github.com/temoTxt/PyPhysics/issues/66](#), [github.com/temoTxt/PyPhysics/pull/70](#). Includes the line-by-line read of DRQM I §III.D Eqs. (III.18)–(III.23) and the FNAL Muon $g - 2$ 2023 cross-particle test (iter-5).

at roughly $57,000 \sigma_{a_\mu}$. It follows that any universal closed form $r/r_0 = f(\alpha)$ in α alone is empirically excluded.

Candidate 2 found that the published expanded K_D of Eq. (III.4) is structurally inadequate to pin r_e via the variational principle⁶. No scale exists where both (a) the NR expansion is valid and (b) the cutoff couples meaningfully to the closure equation. At the electron-radius scale ($\hat{r}_e \sim 1$) the kinetic-energy term dominates by four orders of magnitude; the NR expansion is therefore invalid. At the Bohr scale the expansion is valid, but the cutoff $\hat{r}_e$ is suppressed by α^2 , and the closure matches plain hydrogen to ten significant figures.

Candidate 1 gave the closed form

$$\frac{r_e}{r_0} = \frac{2 - a_e}{2(2 + a_e)} \quad (3)$$

matching the triangulated value to 3.45×10^{-137} . Equation (3) is the algebraic inverse of the framework's existing g -formula (Eq. III.22) evaluated at QED's $a_e(\alpha)$. This corresponds to reproduction-by-inheritance under hypothesis (ii), in which the photon propagator is unchanged and all dual structure is absorbed into the (II.3) Pauli source. We observe that, on this reading, the framework does not independently re-derive Schwinger's vertex result; it inherits it.

The convergent gap across all three candidates is a first-principles derivation of a_e from framework-internal dynamics. Such a derivation requires three framework specifications that the published apparatus does not currently supply. We do not ask the authors to perform the derivation; rather, we ask whether the framework, as intended, specifies (or implicitly fixes) the three pieces below. Each admits a brief answer.

The three questions

1. The proper-time photon propagator $D_F^{(\tau)}(x - y)$

Two natural readings emerge from the framework's existing structure. The first is Schwinger proper-time with b -dispersion,

$$k^2 = (\omega/b)^2 - \mathbf{k}^2, \quad b = \sqrt{c^2 + \mathbf{u}^2}, \quad (4)$$

in which the photon dispersion inherits b -factors from the dual-current structure $J_{\text{Gill}}^\mu = (b\rho, \mathbf{J})$ ⁸. The second is the standard Feynman propagator,

$$k^2 = (\omega/c)^2 - \mathbf{k}^2, \quad (5)$$

⁶Candidate 2: issue #65 and PR #69, github.com/temoTxt/PyPhysics/issues/65, github.com/temoTxt/PyPhysics/pull/69. Includes the iter-4 diagnostic on the NR-expansion / cutoff-scale tension.

⁷Candidate 1: issue #64 and PR #72, github.com/temoTxt/PyPhysics/issues/64, github.com/temoTxt/PyPhysics/pull/72. Closed-form derivation of r_e/r_0 by algebraic inversion of Eq. (III.22) of DRQM I, evaluated at QED's $a_e(\alpha)$.

⁸Dual current $J_{\text{Gill}}^\mu = (b\rho, \mathbf{J})$: Gill-Zachary, *Two Mathematically Equivalent Versions of Maxwell's Equations*, Eq. (12), in [Roadmapping/Converted_Markdown/Two_Mathematically_Equivalent_Versions_of_Maxwells_Equations/](#).

with the b/c conversion absorbed into the source’s coupling rather than into the propagator, and the dual structure entering through modified vertex factors. The two readings give different numerical r_e/r_0 at the same order in α . At the $\log(b/c)$ level they are discriminable at the framework’s precision. It is not clear to us which (if either) is the intended choice.

2. The bound-state propagator $G_C(x, y; E)$ in the (II.3) “potential-in-the-mass” kernel

For the bound-electron one-loop self-energy, one requires a propagator that handles the Coulomb potential non-perturbatively at the cutoff scale, where the NR expansion of Eq. (III.4) is invalid (per Candidate 2’s diagnostic). Standard QED uses the Furry-picture Coulomb-bound propagator, summing the external Coulomb potential to all orders in $Z\alpha$ before perturbing in α . It is not clear from the published apparatus whether the framework specifies (or implicitly uses) a dual analog of the Furry-picture construction, or some other treatment of the bound-state propagator under the (II.3) kernel.

3. The mass-renormalisation prescription at the cutoff

The natural framework-internal closure is

$$\langle K_D \rangle_{r_e} = m_e c^2 + \Delta E_{\text{bind}}^{\text{framework}} + \Delta E_{\text{SE}}^{\text{framework}}. \quad (6)$$

The textbook NR working hypothesis ($\Delta E_{\text{bind}} = \langle V_0 \rangle$, $\Delta E_{\text{SE}} = 0$) gives a trivially-zero closure under the published expanded K_D , per Candidate 2’s diagnostic. We seek the framework’s intended form for $\Delta E_{\text{SE}}^{\text{framework}}(r_e)$ at the cutoff scale. The relevant analog is QED’s on-shell mass renormalisation condition,

$$m_{\text{phys}} = m + \Sigma(p)|_{p^2=m^2c^2}, \quad (7)$$

now to be transposed into the dual framework’s proper-time formulation.

Posture

Any one of these would advance the corresponding candidate; all three together would permit the full hypothesis-(i) calculation. The thread is non-urgent. The triangulated value

$$r_e/r_0 = 0.499\,420\,509\,912\,831\,7 \quad (8)$$

already serves as the campaign’s current best refinement at the framework’s precision floor⁹. It is consistent with the framework’s apparatus at every observable in the joint fit. The substantive content of the three questions is whether the framework, as intended, contains the specifications that would close the gap from empirical cutoff to framework-derived quantity. The alternative is that closing that gap is genuinely new theory work the published apparatus does not yet support.

If the answer to any of (1), (2), (3) is that the framework does not yet specify the relevant piece, that is itself a substantive result. It would mean the campaign’s terminal verdict on

⁹Triangulation note: `Roadmapping/Author_Reports/2026-05_re_triangularization_followup_for_gill.md`.

r_e/r_0 should be recorded as **Branch B, QED-inheritance, pending future framework development**, rather than as **Branch A, from first principles**. Either outcome closes the question honestly; what is at stake here is which honest framing is the right one.

The corresponding responses will be recorded on the framework-specifications issue¹⁰, and the derivation work resumed once the specifications are in hand.

— Trey

References

¹⁰Issue #75 (framework-specifications), github.com/temoTxt/PyPhysics/issues/75.